

UML Diagrams

Add expense sequence

```
sequenceDiagram
    participant U as Member
    participant A as API
    participant S as Expense service
    participant D as PostgreSQL
    U->>A: POST expense
    A->>S: validate and calculate splits
    S->>D: store expense, splits, activity
    D-->>A: expense
    A-->>U: 201 Created

## Balance activity

flowchart TD
    A[Load members, expenses, settlements] --> B[Compute net paise]
    B --> C{Sum equals zero?}
    C -->|yes| D[Sort creditors and debtors]
    D --> E[Match largest amounts]
    E --> F[Return transfers]
    C -->|no| G[Raise invariant error]

## Backend class view

classDiagram
    class GroupService
    class ExpenseService
    class BalanceService
    class DebtSimplifier
    GroupService --> PrismaClient
    ExpenseService --> PrismaClient
    BalanceService --> DebtSimplifier
```